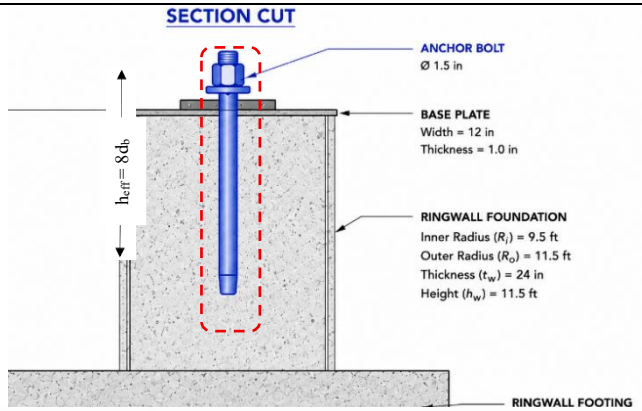
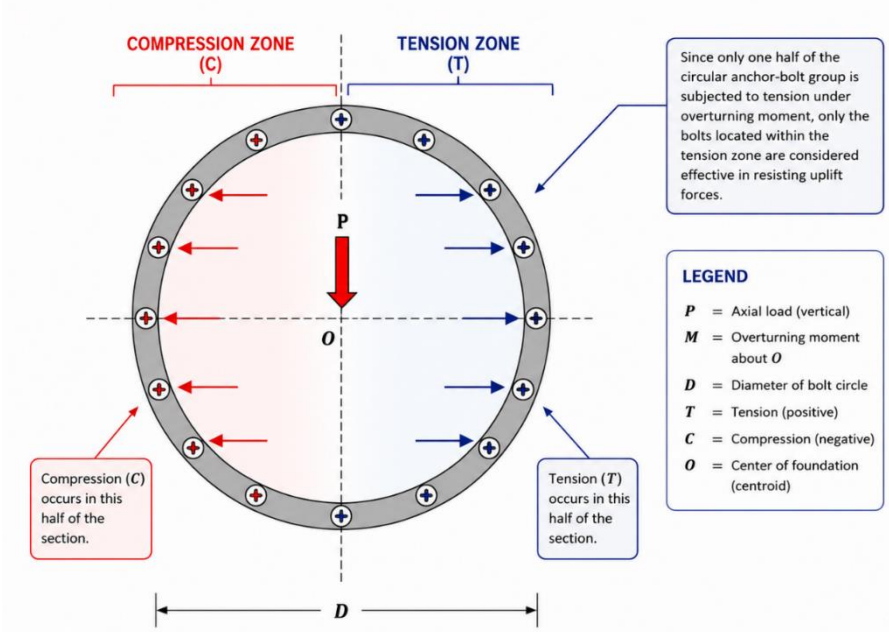
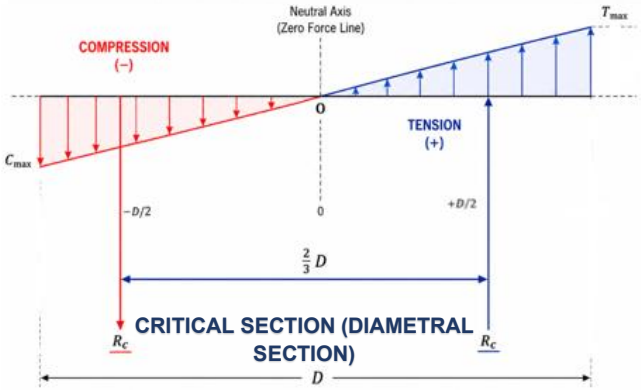
1. Anchor Bolts segment

input

Design Parameter	Symbol	Value	Source
Factored axial load (LRFD)	P_u	1637.5732 kips	Main Tank Design (Predefined)
Factored moment (LRFD)	M_u	5673.437 kip-ft	Main Tank Design (Predefined)
Factored shear force (LRFD)	V_u	57.22 kips	Main Tank Design (Predefined)
Number of anchor bolts	N_β	4	User Input
Diameter circle representing axisymmetric spacing between legs/columns	Dcl	41′-0.5625″	Main Tank Geometry (Predefined)
Anchor bolt diameter	d_β	1.5 in	User Input
Effective embedment depth	h_ef	35 in	User Input
Concrete compressive strength	f'c	4000 psi	User Input
Anchor bolt yield strength	F_γ	36 ksi	User Input
Anchor bolt ultimate tensile strength	F_u	58 ksi	User Input
Edge distance	$e_x = e_\gamma$	13.5 in	AISC Prescribed Minimum
Pedestal dimensions	B × L	39 in × 39 in	User Input
Bolt arrangement dimensions	b × l	12 in × 12 in	From the Pedestal Design
Strength reduction factors (tension, shear)	ϕ_t, ϕ_v	0.75, 0.75	ACI 318-19







Tank radius: $R = 10'6" \; ft$

Governing moment: $M_u = 4847.67 \; kip$

$-\text{ft} ft$,

Inner face: 10.5−1=9.5ft (matches9′−6")

Outer face: 10.5+1=11.5ft

Ring wall thickness = 2 ft

1. Governing Overturning Moment: From calculation

$$M_u = 4847.67 \text{ kip-ft}$$

Diameter of bolt circle:

$$D = 21 \text{ ft}$$

Effective lever arm:

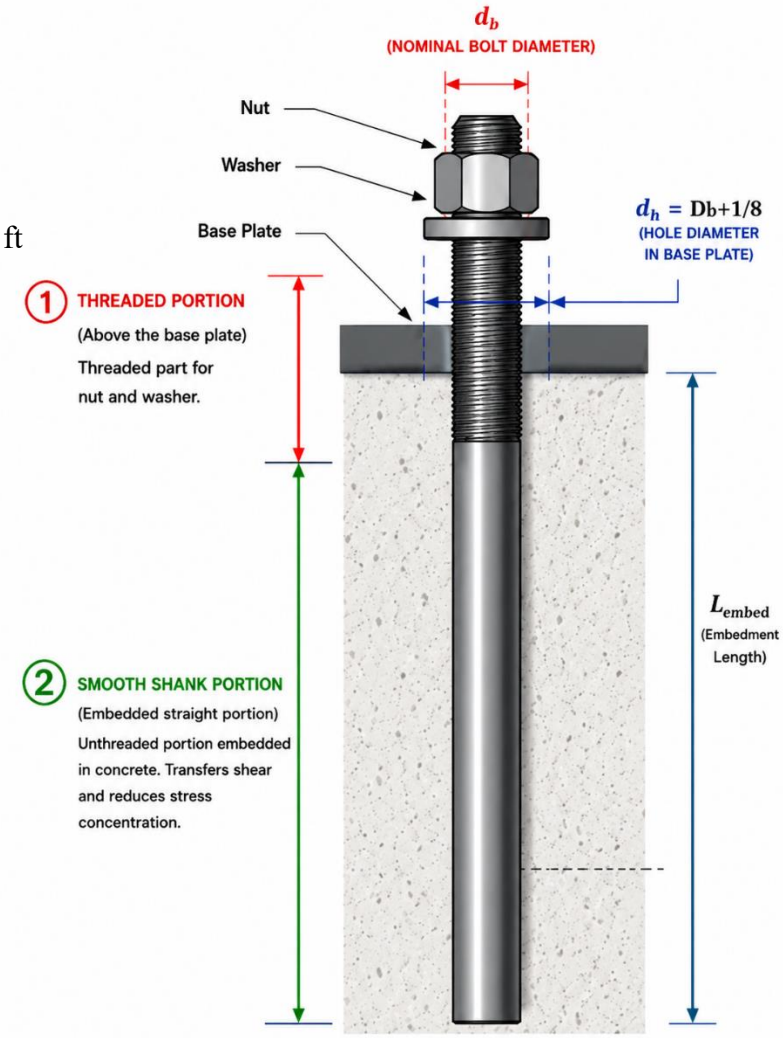
$$0.67D = 0.67(21) = 14.07 \text{ ft}$$

Total uplift force:

$$T_u = \frac{M_u}{0.67D}$$

$$T_u = \frac{4847.67}{14.07}$$

$$T_u = 344.54 \text{ kips}$$



2. Tension Force Per Bolt

You used:

$$T_{bolt} = \frac{2T}{N_b}$$

This is correct for a simplified circular anchor group assumption.

Given:

$N_b = 20$

$$T_{bolt} = \frac{2(344.54)}{20}$$

$$T_{bolt} = 34.45 \text{ kips}$$

3. Bolt Area

We choose reasonable trial values:

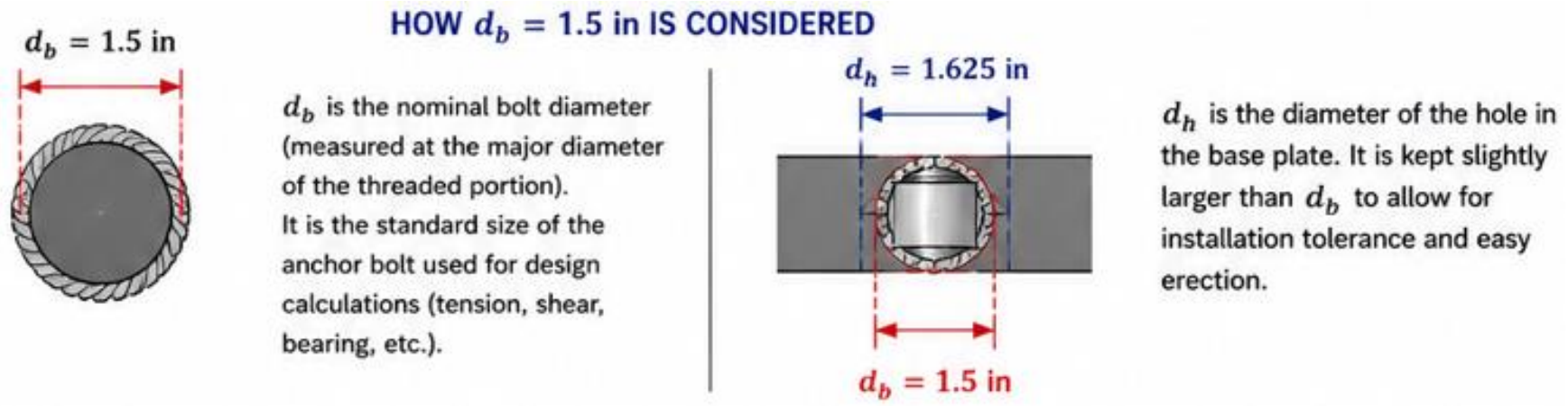
- $N_b = 20$ bolts
- $d_b = 1.5\ in$….. (In accordance with ACI 318 anchor design provisions

the anchor-bolt diameter d_b is taken as the nominal bolt diameter, corresponding to the major diameter of the threaded anchor rod.)

- $d_h = 1.625\ in$ (1/8 + d_b)

- $r_b = 10.75\,ft = 129\,in$
- n= 6 threads/in……(ASME B1.1 — Unified Inch Screw Threads (UN and UNR Thread Form))
- $\alpha_b = 0°$

$$A_b = \frac{\pi d_b^2}{4}$$
$$A_b = \frac{\pi(1.5)^2}{4}$$
$$\boldsymbol{A_b = 1.767\ in^2}$$



Bolt Tensile Stress Area (Recommended)
For threaded anchor rods, ACI 318 and ASTM anchor design practice recommend using the tensile stress area rather than gross shank area because threaded portions govern tensile rupture.
Use:

$$A_t = 0.7854\left(d_b - \frac{0.9743}{n}\right)^2$$

Where:

- A_t= tensile stress area
- d_b= nominal bolt diameter
- n= threads per inch

For 1.5 in diameter anchor rod:

$$A_t \approx 1.41\ in^2$$

Adopt:

$$A_t = 1.41 \text{ in}^2$$

instead of gross area:

$$\boldsymbol{A_b = 1.767 \text{ in}^2}$$

4. Hole Area

$$A_h = \frac{\pi d_h^2}{4}$$

$$A_h = \frac{\pi(1.625)^2}{4}$$

$$\boldsymbol{A_h = 2.074 \text{ in}^2}$$

5. Angular Spacing

$$\Delta\theta_b = \frac{360}{N_b}$$

$$\boldsymbol{\frac{360}{20} = 18^\circ}$$

6. Bolt Coordinates

For bolt i:

$$\theta_i = (i-1) \times 18^\circ$$

$$x_i = r_b \cos\theta_i$$

$$y_i = r_b \sin\theta_i$$

e.g. if i=1 xi=0, Where: $\boldsymbol{r_b = 10.75\ ft}$

If i=2

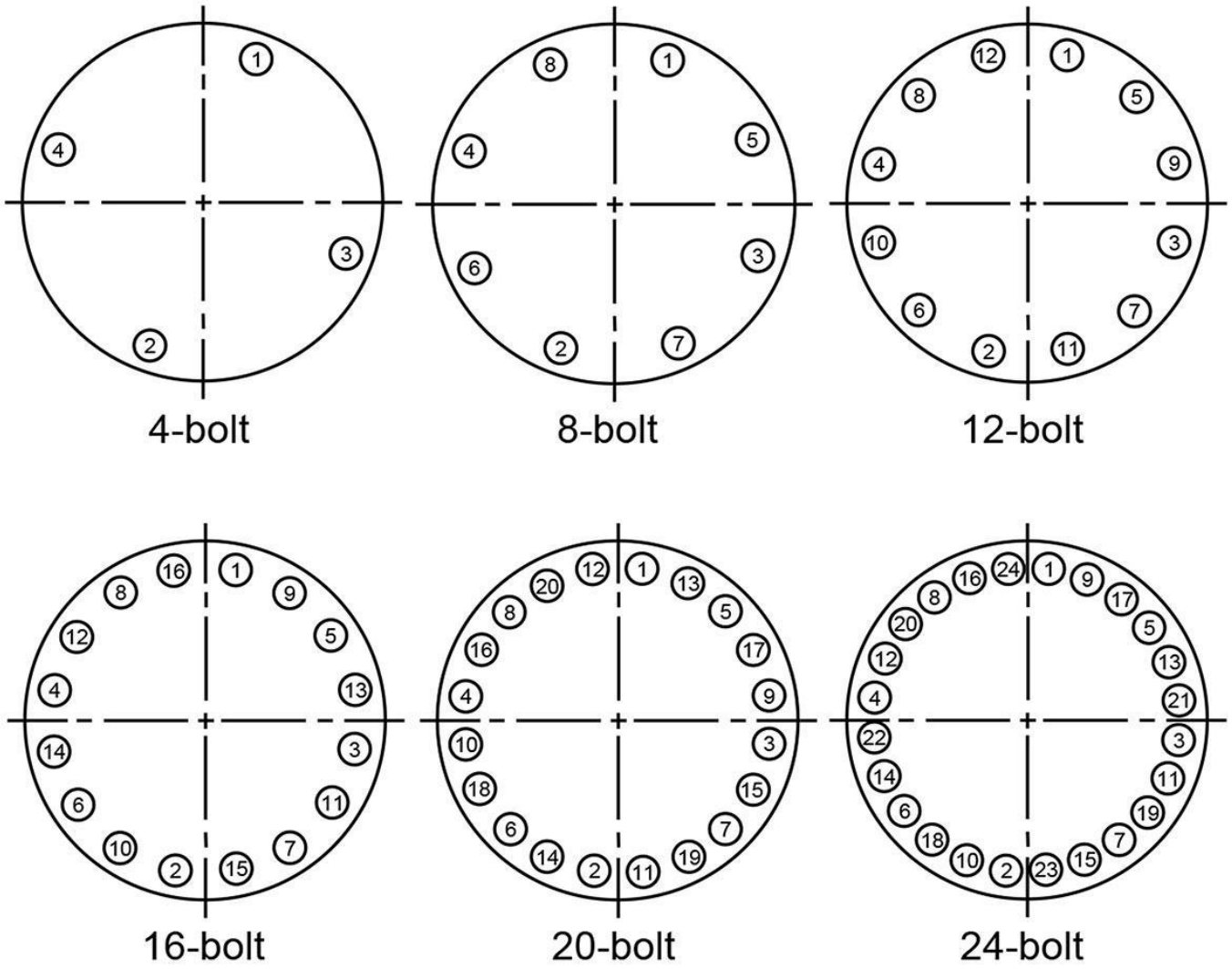
$$\theta_i = (2-1) \times 18^\circ = 18^\circ$$

$$x_2 = 10.75 \text{ ft x} \cos 18^\circ = 10.23\ ft$$

$$y_2 = 10.75 \text{ ft x} \sin 18^\circ = 3.32\ ft$$

Anchor Bolt Layout Table

Bolt	Angle (°)	x (ft)	y (ft)	Location
1	0	10.75	0	Right
2	18	10.23	3.32	Upper-right
3	36	8.7	6.32	Upper-right
4	54	6.32	8.7	Upper-right
5	72	3.32	10.23	Upper-right
6	90	0	10.75	Top
7	108	-3.32	10.23	Upper left
8	126	-6.32	8.7	Upper left
9	144	-8.7	6.32	Upper left
10	162	-10.23	3.32	Upper left
11	180	-10.75	0	Left
12	198	-10.23	-3.32	Lower left
13	216	-8.7	-6.32	Lower left
14	234	-6.32	-8.7	Lower left
15	252	-3.32	-10.23	Lower left
16	270	0	-10.75	Bottom
17	288	3.32	-10.23	Lower-right
18	306	6.32	-8.7	Lower-right
19	324	8.7	-6.32	Lower-right
20	342	10.23	-3.32	Lower-right



7. Arc Spacing

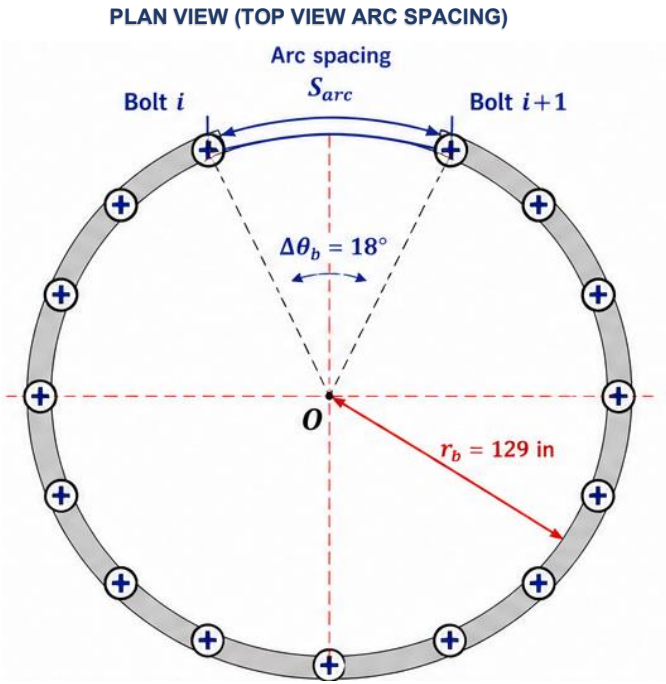
Use radius in inches:

$$r_b = 129 \text{ in}$$

$$S_{arc} = r_b \Delta\theta_b \frac{\pi}{180}$$

$$S_{arc} = 129(18)\frac{\pi}{180}$$

$$S_{arc} = 40.53 \text{ in}$$



8. Chord Spacing

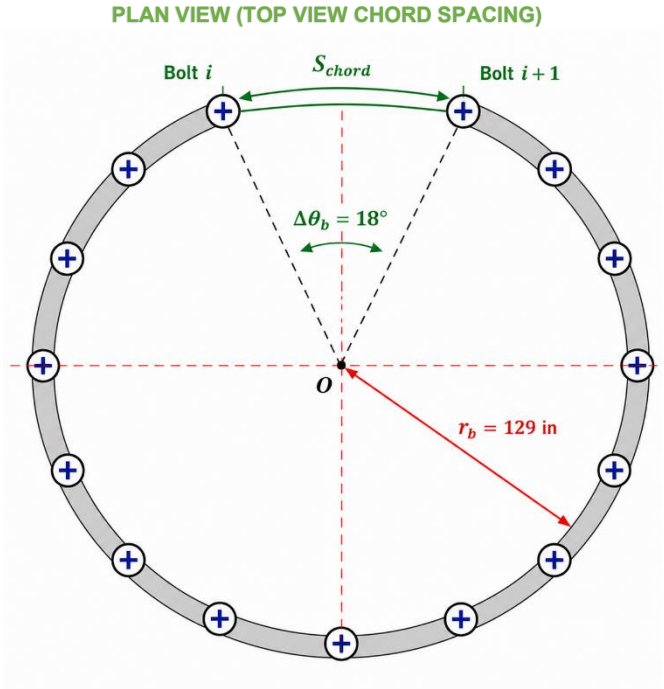
$$S_{chord} = 2r_b \sin\left(\frac{\Delta\theta_b}{2}\right)$$

$$S_{chord} = 2(129)\sin(9°)$$

$$S_{chord} = 40.35 \text{ in}$$

9. Number of Bolts Per Segment

$$n_{bs} = \frac{N_b}{n_s}$$



Typical choices:

- 4 segments (quadrants)
- 6 segments (60° each)
- 8 segments (45° each)

Case 1: 6 Segments

If:

- $N_b = 20$
- $n_s = 6$

$n_{bs} = 3.33\ldots$. Not Acceptable (Software Shouldn't accept this or Flag it).

Case 2: 4 Segments

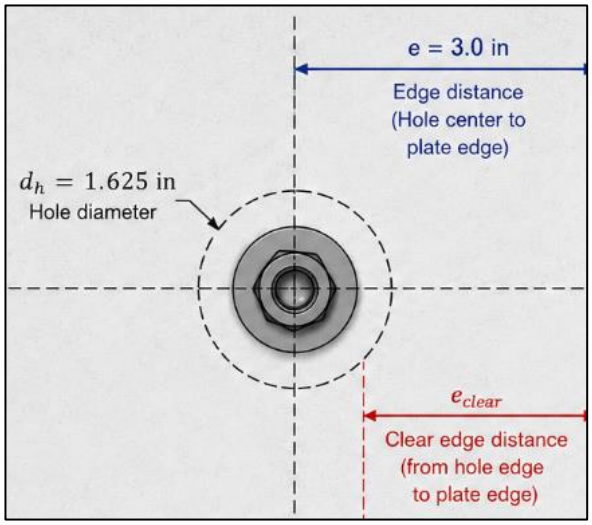
- $n_{bs} = \frac{20}{4} = 5$*.....Acceptable*

Case 3: 5 Segments

- $n_{bs} = \frac{20}{5} = 4$*.....Acceptable*

10. Bolt Clear Edge Distance Check

If the hole is located between plate edges, the minimum clear edge distance can be written as:



$$d_h = 1.625 \text{ in}$$

$$e_{clear} = e - \frac{d_h}{2}$$

$$e_{clear} = 3.0 - \frac{1.625}{2}$$

$$e_{clear} = 3.0 - 0.8125$$

$$e_{clear} = 2.1875 \text{ in}$$

$$\boldsymbol{e_{clear} = 2.19 \text{ in}}$$

Check Against Code

From ACI / AISC / practical detailing:

Typical minimums:

- ≥ 1.5 in (absolute minimum)
- ≥ 1.5 × bolt diameter (preferred)

For your bolt:

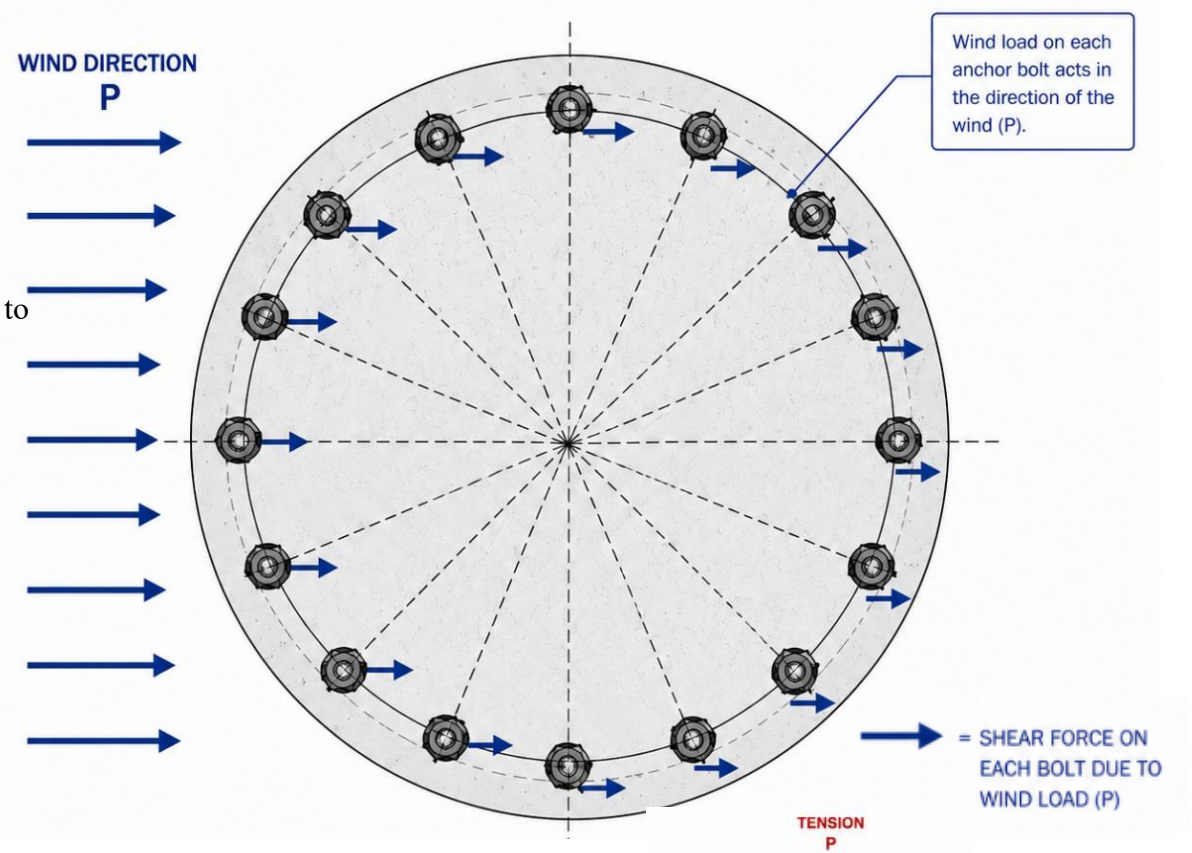
$$1.5 \times d_b = 1.5 \times 1.5 = 2.25\ in$$

e_{clear}*=2.19in>1.5in……..*✓***SAFE***

Recommend: Increase edge distance to **3.25–3.5 in**

11. Shear Per Bolt
From your data: Wind shear ≈ **20.683 kips**

Each Bolt Resists Shear: All anchor bolts are subjected to shear force parallel to the wind direction (P).



Per bolt:

$$V_{bolt} = \frac{20.683}{20}$$

$$\boldsymbol{V_{bolt} = 1.034\ kips}$$

12. Tensile Capacity

Using:

$$\boldsymbol{\phi P_{nt} = \phi A_b F_u}$$

Given:

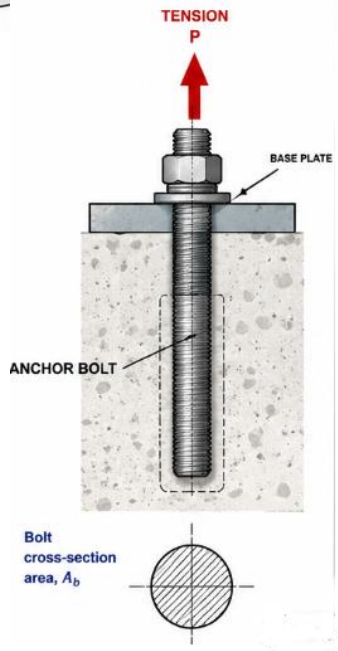
$\phi = 0.75$

$A_b = 1.767$

$F_u = 58$ ksi

$$\phi P_{nt} = 0.75(1.767)(58)$$

$$\boldsymbol{\phi P_{nt} = 76.87\ kips}$$



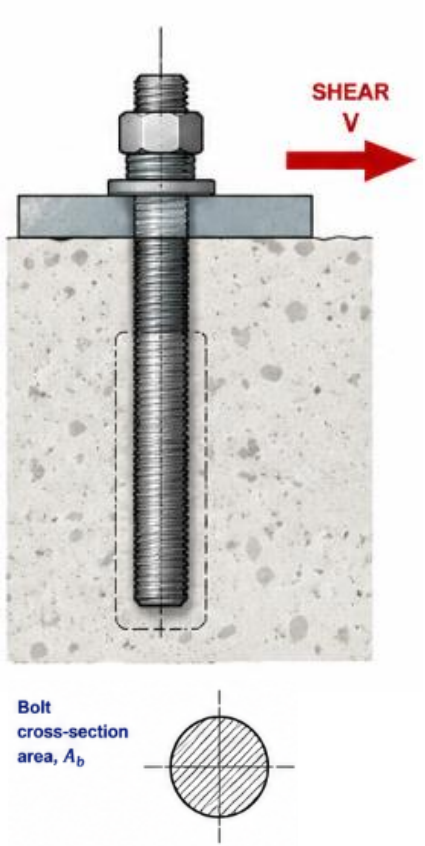
13. Shear Capacity

$$F_{nv} = 0.6F_u$$

$$F_{nv} = 0.6(58) = 34.8 \text{ ksi}$$

$$\phi V_n = 0.75(1.767)(34.8)$$

$$\boldsymbol{\phi V_n = 46.11 \text{ kips}}$$



14. Interaction Check

$$\text{Utilization} = \left(\frac{Nu}{\phi N_n}\right)^{\alpha} + \frac{V_u}{\phi V_n} \leq 1 \text{.....} \textit{ACI 318-19 Sec17.6}$$

$$\text{Utilization} = \frac{T_{bolt}}{\phi P_{nt}} + \frac{V_{bolt}}{\phi V_n} \leq 1$$

$$= \frac{34.45}{76.87} + \frac{1.034}{46.11}$$

$$= 0.448 + 0.022$$

$$\boldsymbol{=0.470}$$

$$\boldsymbol{Utilization = 0.47 < 1.0 \Rightarrow SAFE}$$

15. Anchor Bolt Length Design:

- **Minimum embedment (code-based)**

ACI / AISC / industry:

$$h_{ef} \geq 8d_b$$

$$h_{ef} = 8 \times 1.5 = 12\ in$$

16. Pull Out Test/ Concrete Breakout Capacity

$$N_b = k_c \sqrt{f_c'}\, h_{ef}^{1.5}$$

where:

- N_b = *concrete breakout strength*
- k_c = *empirical constant*
- f_c' = *concrete compressive strength*
- h_{ef} = *effective embedment depth.*
- *Embedment = approximately 9–10 ft*
- *Effective embedment = assume* $h_{ef} = 108$ *in conservative*

Based on the foundation detail drawing,

$$h_{ef} = 96 \ in$$

Substituting:

$$N_b = 24\sqrt{4000}(96)^{1.5} = 24(63.25)(940.6)$$
$$N_b = 1{,}427{,}470 \ lb$$
$$N_b \approx 1427 \ kips$$

Using the ACI strength reduction factor:

$$\phi = 0.75$$
$$\phi N_b = 0.75(1427)$$
$$\phi N_b \approx 1070 \ kips$$

Maximum tensile demand per anchor bolt:

$$T_{bolt} = 34.45 \ kips$$

17. Concrete breakout utilization:

$$Utilization = \frac{T_{bolt}}{\phi N_b}$$
$$= \frac{34.45}{1070}$$
$$= 0.032$$
$$0.032 < 1.0$$

-

Parameter	Value
Number of anchor bolts, (N_b)	20
Anchor bolt diameter, (d_b)	1.5 in
Hole diameter, (d_h)	1.625 in
Bolt circle radius, (r_b)	10.75 ft (129 in)
Angular spacing, ($\Delta\theta_b$)	18°
Arc spacing, (S_{arc})	40.53 in
Chord spacing, (S_{chord})	40.35 in
Clear edge distance, (e_{clear})	2.19 in
Bolt tensile area, (A_b)	1.767 in²
Bolt hole area, (A_h)	2.074 in²
Tension demand per bolt, (T_{bolt})	34.45 kips
Shear demand per bolt, (V_{bolt})	1.03 kips
Tensile capacity, ϕP_{nt}	76.87 kips
Shear capacity, ϕVn	46.11 kips
Interaction utilization ratio	0.47
Governing interaction equation	Utilization $= \frac{T_{bolt}}{\phi P_{nt}} + \frac{V_{bolt}}{\phi V_n} \leq 1$
Governing design check	Combined tension and shear
Concrete breakout check	Not governing
Anchor embedment behavior	Deep cast-in plate-anchor system
Final anchor bolt design status	✓ SAFE

Note: The present calculations primarily evaluate steel anchorage strength and preliminary anchorage behavior. Final design should also verify reinforcement detailing, pedestal confinement, local concrete stresses, and constructability requirements in accordance with ACI 318 Chapter 17 and project-specific structural drawings.

Anchor Bolt with Plate Attached (Concrete Breakout Check)**Anchor Bolt Size (in)**	**Quantity**	**Factored Tension Load (kip)**	**Effective Embedment (in)**
1.50	20	34.45	96

